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Anul I

Examen parțial Analiză

1. $\sum_{n=1}^{\infty} \frac{2 \cdot 5 \cdot 8 \cdot \ldots \cdot (3n-1)}{1 \cdot 5 \cdot 9 \cdot \ldots \cdot (4n-3)} \cdot \frac{1}{2^n \cdot (3n+1)}$

$u_n = \frac{2 \cdot 5 \cdot 8 \cdot \ldots \cdot (3n-1)}{1 \cdot 5 \cdot 9 \cdot \ldots \cdot (4n-3)} \cdot \frac{1}{2^n (3n+1)}$

Aplic criteriul raportului:

$\lim_{n \to \infty} \frac{u_{n+1}}{u_n} = \lim_{n \to \infty} \frac{2 \cdot 5 \cdot 8 \cdot \ldots \cdot (3n+2)}{1 \cdot 5 \cdot 9 \cdot \ldots \cdot (4n+1)} \cdot \frac{1}{2^{n+1}(3n+4)} \cdot$

$\cdot \frac{1 \cdot 5 \cdot 9 \cdot \ldots \cdot (4n-3)}{2 \cdot 5 \cdot 8 \cdot \ldots \cdot (3n-1)} \cdot \frac{2^n (3n+1)}{1} =$

$= \lim_{n \to \infty} \left(\frac{3n+2}{4n+1} \cdot \frac{1}{2} \cdot \frac{3n+1}{3n+4} \right) = \lim_{n \to \infty} \left[\frac{n(3+\frac{2}{n})}{n(4+\frac{1}{n})} \cdot \frac{1}{2} \cdot \frac{n(3+\frac{1}{n})}{n(3+\frac{4}{n})} \right] =$

$= \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{3}{3} = \frac{3}{8} < 1 \Rightarrow$ conform criteriului raportului seria $\sum_{n=1}^{\infty} u_n$ este convergentă

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2. $\sum_{n\geq 1}(-1)^n \frac{\sqrt{n^{19}+20n^a+18}-n^9\cdot\sqrt{n}}{n^2}$, $a\in(0,19)$

$u_n=(-1)^n \frac{\sqrt{n^{19}+20n^a+18}-n^9\cdot\sqrt{n}}{n^2}$,

Studiez absolut convergența $\Rightarrow$

$$\sum_{n\geq 1}\left|(-1)^n \frac{\sqrt{n^{19}+20n^a+18}-n^9\sqrt{n}}{n^2}\right| =$$

$$= \sum_{n\geq 1} \frac{\sqrt{n^{19}+20n^a+18}-n^9\sqrt{n}}{n^2} =$$

$$= \sum_{n\geq 1} \frac{n^{19}+20n^a+18-n^{19}}{n^2\left(\sqrt{n^{19}+20n^a+18}+n^9\sqrt{n}\right)} =$$

$$= \sum_{n\geq 1} \frac{20n^a+18}{n^2\left(\sqrt{n^{19}+20n^a+18}+n^9\sqrt{n}\right)} \Rightarrow$$

$$\Rightarrow \sum_{n\geq 1}|u_n| \sim \sum_{n\geq 1}\frac{n^a}{n^{23/2}}$$

$$\sum_{n\geq 1}\frac{n^a}{n^{23/2}} = \sum_{n\geq 1}\frac{1}{n^{23/2-a}}$$ (Serie Riemann)

Dacă $\frac{23}{2}-a>1 \Leftrightarrow a<\frac{23}{2}-1 \Leftrightarrow a<\frac{21}{2}$

$\Leftrightarrow a\in\left(0,\frac{21}{2}\right)$ seria $\sum_{n\geq 1}\frac{1}{n^{23/2-a}}$ este convergentă, deci din cel de-al treilea criteriu de comparație

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și $\sum_{n\geq 1} |u_n|$ este convergentă $\Rightarrow \sum_{n\geq 1} u_n$ este absolut convergentă

Dacă $\frac{23}{2} - a \leq 1 \Leftrightarrow a \geq \frac{21}{2} \Leftrightarrow a \in [\frac{21}{2}, 19)$

seria $\sum_{n\geq 1} \frac{1}{n^{23/2-a}}$ divergentă, deci din cel de-al treilea criteriu de comparație și $\sum_{n\geq 1} |u_n|$ este divergentă $\Rightarrow \sum_{n\geq 1} u_n$ nu este absolut convergentă

Studiem acum convergența

Știm acum că $\sum_{n\geq 1} u_n$ este absolut convergentă pentru $a \in (0, \frac{21}{2})$, de unde rezultă că ea este și convergentă pentru $a \in (0, \frac{21}{2})$

Studiem deci convergența pentru $a \in [\frac{21}{2}, 19)$

$$\alpha_n = \frac{\sqrt{n^{19}+20n^a+18} - n^9\cdot\sqrt{n}}{n^2} = \frac{20n^a+18}{n^2\left(\sqrt{n^{19}+20n^a+18}+n^9\sqrt{n}\right)}$$

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$$\lim_{n\to+\infty} a_n = \lim_{n\to+\infty} \frac{20n^a + 18}{n^2(\sqrt{n^{19}} + 20n^a + 18 + n^2\sqrt{n}} =$$

$$= \begin{cases} 0, & a \in [\frac{21}{2}, \frac{23}{2}) \\ 20, & a = \frac{23}{2} \\ +\infty, & a \in (\frac{23}{2}, 19) \end{cases}$$

$\Rightarrow$ Pt $a \in [\frac{21}{2}, \frac{23}{2})$ $\lim_{n\to+\infty} a_n = 0$ și șirul a_n este descrescător, deci conform criteriului lui Leibniz seria este convergentă

$\Rightarrow$ Pt $a \in [\frac{23}{2}, 19)$ șirul termenului general nu are limită, deci conform criteriului necesar seria nu este convergentă

3. $f_n : \mathbb{R} \to \mathbb{R}$, $f_n(x) = \frac{nx}{1+n^6x^2}$

Studiez convergența simplă sau punctuală

$$\lim_{n\to+\infty} f_n(x) = \lim_{n\to+\infty} \frac{nx}{1+n^6x^2} = 0 \Rightarrow f_n \xrightarrow{p} 0$$

$f(x) = 0$

Studiez acum convergența uniformă:

$$\lim_{n\to+\infty} \left(\sup_{x\in\mathbb{R}} |f_n(x) - f(x)| \right) = \lim_{n\to+\infty} \left(\sup_{x\in\mathbb{R}} |f_n(x)| \right)$$

Fie $g(x) = \frac{nx}{1+n^6x^2}$

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$$g'(x) = \frac{n(1+n^6x^2) - nx(2n^6x)}{(1+n^6x^2)^2} =$$

$$= \frac{n + n^7x^2 - 2n^7x^2}{(1+n^6x^2)^2} = \frac{n - n^7x^2}{(1+n^6x^2)^2}$$

$g'(x) = 0 \iff n - n^7x^2 = 0 \Rightarrow x^2 = \frac{n}{n^7} \iff$

$\iff x^2 = \frac{1}{n^6} \iff x_{1/2} = \pm \frac{1}{n^3}$

x	$-\infty$	$-1/n^3$	$1/n^3$	$+\infty$
$g'(x)$		0	0	
$g(x)$				

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4. $\sum_{n\geq 2} \frac{(-1)^n}{n} \cdot x^{2n}$, $x \in \mathbb{R}$.

$$a_n = \frac{(-1)^n}{n}$$

$$R = \lim_{n\to\infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n\to\infty} \left| \frac{(-1)^n}{n} \cdot \frac{n+1}{(-1)^{n+1}} \right| = 1 \Rightarrow$$

$\Rightarrow$ Intervalul de convergență este $(-1, 1)$

Trebuie să studiem convergența în capete, dar observăm că dacă $x_1 = -1 \Rightarrow (x_1)^{2n} = 1$ și dacă $x_2 = 1 \Rightarrow x_2^2 = 1$

$\Leftrightarrow x_1^2 = x_2^2 = 1$, deci e suficient să studiem convergența într-unul din capete

$x = 1 \Rightarrow$ seria devine $\sum_{n\geq 2} (-1)^n \cdot \frac{1}{n}$

$$u_n = \frac{1}{n}$$

$$\frac{u_{n+1}}{u_n} = \frac{1}{n+1} \cdot \frac{n}{1} = \frac{n}{n+1} < 1 \Rightarrow \text{șirul } u_n \text{ este descrescător}$$

$$\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{1}{n} = 0$$

$\Rightarrow$ Cum șirul u_n este descrescător și tinde la 0 $\Rightarrow$

$\Rightarrow$ conform criteriului lui Leibniz că $\sum_{n\geq 2} (-1)^n \cdot \frac{1}{n}$ este convergentă $\Rightarrow$ Mulțimea de convergență este $[-1, 1]$

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$S(x) = ?$

$$\frac{1}{1-x} = \sum_{n \geq 0} x^n$$

$$\frac{1}{1-(-x^2)} = \sum_{n \geq 0} (-x^2)^n = \sum_{n \geq 0} (-1)^n \cdot x^{2n}$$

$$\Rightarrow \frac{1}{1+x^2} = 1 - x^2 + \sum_{n \geq 2} (-1)^n \cdot x^{2n} \Rightarrow$$

$$\Rightarrow \sum_{n \geq 2} (-1)^n \cdot x^{2n} = \frac{1}{1+x^2} \overset{1+x^2}{-1} + \frac{1+x^2}{x^2} \Rightarrow$$

$$\Rightarrow \sum_{n \geq 2} (-1)^n \cdot x^{2n} = \frac{1 - 1 - x^2 + x^2 + x^4}{1+x^2} \Rightarrow$$

$$\Rightarrow \sum_{n \geq 2} (-1)^n \cdot x^{2n} = \frac{x^4}{1+x^2} \quad | : x \Rightarrow$$

$$\Rightarrow \sum_{n \geq 2} (-1)^n \cdot x^{2n-1} = \frac{x^3}{1+x^2} \quad | \text{ Integrez ambii membri}$$

$$\Rightarrow \sum_{n \geq 2} (-1)^n \cdot \frac{x^{2n}}{2n} = \int \frac{x^3}{1+x^2} dx$$

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$$\int \frac{x^3}{1+x^2}\,dx = \int x \cdot \frac{x^2-1+1}{1+x^2}\,dx =$$

$$= \int x \cdot \frac{1+x^2}{1+x^2}\,dx - \int \frac{x}{1+x^2}\,dx =$$

$$= x^2 - \frac{1}{2}\int \frac{2x}{1+x^2}\,dx = x^2 - \frac{1}{2}\int \frac{u'(x)}{u(x)}\,dx =$$

$$= x^2 - \frac{1}{2}\ln(1+x^2) + C$$

$$x=0 \Rightarrow C=0$$

$$\Rightarrow \sum_{n\geq 2} (-1)^n \cdot \frac{x^{2n}}{2n} = x^2 - \frac{1}{2}\ln(1+x^2) \Rightarrow$$

$$\Rightarrow \sum_{n\geq 2} \frac{(-1)^n}{n} \cdot x^{2n} = 2x^2 - \ln(1+x^2) \Rightarrow$$

$$\Rightarrow S(x) = 2x^2 - \ln(1+x^2)$$

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5. $f: \mathbb{R}^3 \to \mathbb{R}^2$, $f(x,y,z) = (\sqrt{4y^2+z^2+1}, \ln(x^2+3z^4+2))$

$$df = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z}$$

$u(x,y,z) = \sqrt{4y^2+z^2+1}$; $v(x,y,z) = \ln(x^2+3z^4+2)$

$$\frac{\partial u}{\partial x} = 0 \; ; \; \frac{\partial u}{\partial y} = \frac{4y}{\sqrt{4y^2+z^2+1}} \; ; \; \frac{\partial u}{\partial z} = \frac{z}{\sqrt{4y^2+z^2+1}}$$

$$\frac{\partial v}{\partial x} = \frac{2x}{x^2+3z^4+2} \; ; \; \frac{\partial v}{\partial y} = 0 \; ; \; \frac{\partial v}{\partial z} = \frac{12z^3}{x^2+3z^4+2}$$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x} = 0 + \frac{2x}{x^2+3z^4+2} \frac{\partial f}{\partial v} =$$

$$= \frac{2x}{x^2+3z^4+2} \frac{\partial f}{\partial v}$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial y} = \frac{4y}{\sqrt{4y^2+z^2+1}} \frac{\partial f}{\partial u}$$

$$\frac{\partial f}{\partial z} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial z} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial z} = \frac{z}{\sqrt{4y^2+z^2+1}} \frac{\partial f}{\partial u} +$$

$$+ \frac{12z^3}{x^2+3z^4+2} \frac{\partial f}{\partial v}$$

$$\Rightarrow df = \frac{2x}{x^2+3z^4+2} \frac{\partial f}{\partial v} + \frac{4y}{\sqrt{4y^2+z^2+1}} \cdot \frac{\partial f}{\partial u} + \frac{z}{\sqrt{4y^2+z^2+1}} \frac{\partial f}{\partial u} +$$

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$$+\frac{12\cdot 23}{x^2+32y+2}\frac{\partial f}{\partial v}$$

$$df(1,1,0)=\frac{2}{3}\frac{\partial f}{\partial v}+\frac{4}{\sqrt{5}}\frac{\partial f}{\partial u}+0+0=\frac{2}{3}\frac{\partial f}{\partial v}+\frac{4}{\sqrt{5}}\frac{\partial f}{\partial u}$$

6. $E=\frac{\partial^2 f}{\partial x^2}+\frac{\partial^2 f}{\partial y\partial x}$

$f(u,v)$; $u=4x^2+1$, $v=x(2+y)$ și f este funcție de clasă C^2 ;

$$\frac{\partial u}{\partial x}=12x^2;\ \frac{\partial u}{\partial y}=0;\ \frac{\partial v}{\partial x}=2+y\ \frac{\partial v}{\partial y}=x$$

$$\frac{\partial f}{\partial x}=\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x}=12x^2\cdot\frac{\partial f}{\partial u}+2\frac{\partial f}{\partial v}$$

$$\frac{\partial^2 f}{\partial x^2}=\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)=\frac{\partial}{\partial x}\left(12x^2\cdot\frac{\partial f}{\partial u}+2\frac{\partial f}{\partial v}\right)=$$

$$=24x\frac{\partial f}{\partial u}+12x^2\frac{\partial f}{\partial u}\left(\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x}\right)+$$

$$+2\frac{\partial f}{\partial v}\left(\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x}\right)=$$

$$=24x\frac{\partial f}{\partial u}+144x^4\cdot\frac{\partial^2 f}{\partial u^2}+24x^2\frac{\partial^2 f}{\partial u\partial v}+$$

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$$+ 24x^2 \frac{\partial^2 f}{\partial u \partial v} + \cancel{24} + 4 \frac{\partial^2 f}{\partial v^2} =$$

$$= 24x \frac{\partial f}{\partial u} + 144 v^4 \frac{\partial^2 f}{\partial u^2} + 48x^2 \frac{\partial^2 f}{\partial u \partial v} + 4 \frac{\partial^2 f}{\partial v^2}$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = 0$$

$$\Rightarrow E = \frac{\partial^2 f}{\partial x^2}.$$

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